Recommendations for US Caribbean Fisheries Stock Assessment and Management Alternatives

Summary

A primary goal of the Southeast Fisheries Science Center's (SEFSC) Caribbean Strategic Planning project is to determine data appropriate methods to inform management. The first step in accomplishing that goal to support the management bodies in the region is to identify, investigate, and test alternative management methods. A working group was established to understand the literature and approaches used in other regions, summarize existing needs of management agencies associated with alternative management methods and socialize the suite of alternative methods with the SSC, APs, GC, Council and any other relevant bodies, match agency needs with the alternative management methods and understand what is possible today and make recommendations as to what data gap filling would be most beneficial given the suite of strategies available to inform management, and summarize and distribute information on appropriate alternative management strategies and approaches in a format that is readily available and used by the CFMC and SSC and provide improved communication approaches that builds public support for use of these approaches. This document contains those findings.

A deliverable of the Caribbean Strategic Planning Project

Updated June 2026

Table of Contents

Acronyms

Acronym	*Definition*
CMFC	Caribbean Fishery Management Council
CIMAS	Cooperative Institute for Marine and Atmospheric Studies
CSP	Caribbean Strategic Planning
EAFM	Ecosystem Approach to Fisheries Management
EBFM	Ecosystem Based Fisheries Management
ERAEF	Ecological Risk Assessment for the Effects of Fishing
LANTERN	Leveraging Abilities, Needs, Talents, Energies and Resources Network
MSE	Management Strategy Evaluation
NEFSC	Northeast Fisheries Science Center
NOAA	National Oceanic and Atmospheric Administration
NMFS	National Marine Fisheries Service
PR	Puerto Rico
PR DRNA	Puerto Rico Departamento de Recursos Naturales y Ambientales
SEDAR	Southeast Data Assessment and Review
SEFSC	Southeast Fisheries Science Center
SERO	Southeast Regional Office
SPR	Spawning Potential Ratio
UPR	University of Puerto Rico
USVI	United States Virgin Islands
USVI DFW	United States Virgin Islands Division of Fish and Wildlife

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Introduction

Due to major data limitations, traditional approaches have seen limited success, and there is a large appetite for innovation, particularly to move beyond the limitations of this region for classical single-species approaches.

Participants volunteered to join this strategic planning working group. During the initial meeting of the group, the participants developed their purpose, goals, and ideal outcomes, which are documented in the Team Charter. The group met 17 times over the course of three years to accomplish the goals. Section A contains those goals and what was accomplished under each goal.

A. Goals & Summary of Actions to Address Goals:

1. To build on the literature review of analytical methods used in other regions (including methods that may have been previously deemed as not aligned with legal requirements) and classify and annotate the analytical methods, potentially as part of a student project and including engagement with external technical experts and stakeholders to the extent possible.

2. To summarize the types of methods that are allowable under MSA rules and also what we can accomplish based on available data and existing needs of management agencies and socialize the suite of alternative methods with the SSC, APs, GC, Council and any other relevant bodies. This should be in conjunction with SERO and SEFSC.

3. To match agency needs with the alternative management methods and understand what is possible today and make recommendations as to what data gap filling would be most beneficial, given the suite of strategies available to inform management. Review the SEFSC Data Triage and Strategic Planning Project to accomplish this goal.

4. To summarize and distribute information on appropriate alternative management strategies and approaches in a format that is readily available and used by the CFMC and SSC and provide improved communication approaches that build public support for use of these approaches.

Peer-reviewed publications and agency reports were compiled to understand other available approaches to fisheries management (Table 1). Additionally, the NOAA Fisheries Integrated Toolbox contains a variety of publicly accessible software tools for analyses regarding resource management.

To achieve goals 2, 3, and 4, the working group first employed the use of a NOAA LANTERN detail to assist in advancing key ecosystem science and management needs for the U.S. Caribbean. Specifically, the selected LANTERN incumbent synthesized potential regional management approaches that match data availability and management needs and explored a feasible ecosystem-based approach. An occupancy modeling framework was tested for use in the U.S. Caribbean using simulated data. This approach may provide an advantage in providing a less biased picture than a traditional single species assessment.

To continue the work of identifying and testing additional alternative management methods, the working group proposed to the CFMC that the next Caribbean SEDAR be a procedural workshop to identify and test potential analytical methods to provide management advice for the federally managed species of the U.S. Caribbean. The workshop was approved and scheduled for August, 2026. Follow SEDAR 103 for updates on this effort.

B. NOAA LANTERN:

LANTERN detail description

The Science Advisor will work with and assist the Southeast Fisheries Science Center's Sustainable Fisheries Division and its collaborators in advancing key ecosystem science and management needs for the U.S. Caribbean Region. The Science Advisor will synthesize potential regional management approaches that match data availability and management needs. This effort is part of a wider U.S. Caribbean Region Strategic Planning effort. It will involve collaboration across multiple working groups and coordination with the Southeast Regional Office, territorial, commonwealth, academic, and private sector partners. A large part of the position will be to coordinate with groups focused on improving data collection to inform assessments and management in the region and harmonize the ongoing efforts to explore feasible ecosystem-based fishery management approaches in the region. This is an exciting opportunity to influence how assessments are conducted and to set up the U.S. Caribbean for future success in fisheries management. Due to major data limitations, traditional approaches have seen limited success, and there is a large appetite for innovation, particularly to move beyond the limitations of this region for classical single-species approaches. With this LANTERN detail, the U.S. Caribbean is your mangrove oyster!

What You'll Do: The Science Advisor will develop executive core qualifications in Leading Change, Results Driven, and Building Coalitions. They will be responsible for the following duties: a) Building on an existing literature review of data-limited management approaches (both single-species and ecosystem-based approaches); b) Classifying and synthesizing assessment and management approaches, and socializing the suite of alternative methods with the Caribbean Fisheries Management Council and its advisory bodies, General Counsel, and any other relevant bodies; c) Attending regular Strategic Planning work group meetings, and integrating with other components of Strategic Planning, such as data triage working groups; d) Supporting the prioritization process of alternative management approaches that are feasible at the present day, and making recommendations for expanded data collection efforts that allow the application of future state approaches; e) Developing and writing white papers and other manuscripts accordingly; and f) Mentoring students from the region to assist in the project (should funding for student assistantship become available)

What you'll learn: The Science Advisor will gain the additional following experience: a) Leading a complex project linked to many distinct and sometimes disparate efforts and numerous collaborator groups; b) Navigating regional demands, national priorities, and developing consensus across groups of people with diverse backgrounds and experiences; and c) Motivating, organizing, and leading people to achieve organizational objectives

Modeling

The LANTERN incumbent synthesized potential regional management approaches that match data availability and management needs and explored a feasible ecosystem-based approach, occupancy modeling. Site occupancy models are often used to study wildlife populations through the use of passive monitoring (e.g., automated camera array) to detect species presence or absence (Goldstein et al., 2024). The LANTERN incumbent tested an occupancy modeling framework for use in the U.S. Caribbean using simulated data. The simulation model was built using R and linked to Rpresence for scenario testing. In this model, habitat was defined as a function of depth between 20 and 40 m, but this could be expanded to other factors like temperature or dissolved oxygen. The known population size was selected as 40,000 fish and placed randomly in the habitat. In the 12-month simulation, the modeling timestamp was selected as daily with monthly mixes. The monthly survival was selected as 0.6 and the detection probability was selected as 0.1 and included a random recruitment event of 20,000 fish halfway through. Finally, 50 cameras were randomly placed throughout the simulation. This base model was tested (Figure 1) and compared to other scenario testing in which one of the model inputs were changed with each simulation including lower detection probability, lower starting abundance, higher starting abundance, higher survival, lower survival, fewer cameras, and abundance as a function of season.

Results

It was found that the model will not work for highly abundant fish (Figure 2) and is too sensitive for low abundance (Figure 3), meaning it would not be useful for management in those scenarios because population changes were not detected. Additionally, in the higher survival scenario (Figure 4), the model was unable to estimate occupancy. However, in the other scenarios (Figures 5-8), the model was successfully able to estimate fish abundance by month as well as extirpation and colonization. Another key point to note is that changes in population would also be detected by fewer cameras, which may be more cost-effective depending on the area to monitor. Daily measurements were selected as the timestamp to mimic a real life wildlife camera, but it would be expected that longer years with quarterly measurements would lead to similar results. The presence of other species stocks could also be built into the model to test multispecies populations or interactions with other species. In the wild, the cameras would be placed where the fish are caught in the preferred habitat or many preferred habitats, unless the movement is dynamic and the interest is tracking that movement seasonally. The modeling framework corrects for a fixed detection probability in the event of low visibility or other factors that would prevent a detection, unlike CPUE which does not account for catchability. The model assumes that the wildlife camera array takes photos at their fixed site every day. If cameras were deployed on gear at different times and locations, rather than being fixed underwater, the model would need to be adjusted. However, a quarterly model with only 10 measurements per quarter could be used to develop detection probabilities. The main takeaway was that for a minimal investment of 50 cameras scattered throughout an important habitat, occupancy modeling could be used to effectively provide informative, ecosystem-level management advice. This approach may provide an advantage in providing a less biased picture than a traditional single species assessment.

C. SEDAR 103:

Workshop objective

The objective of this workshop is to review and discuss alternative approaches to single species assessments as outlined in the literature or as applied in other data-limited regions, and explore ecosystem-based fishery management approaches that have not yet been applied or published. These alternative approaches should be reviewed for potential application with respect to data availability in the region as well as compliance with legal mandates. The workshop should result in research recommendations to advance the development and implementation of alternative assessment methods and provide recommendations on methods that could most easily be applied given the present scenario.

Problem description

The U.S. Caribbean region faces a number of challenges regarding the application of traditional single-species stock assessment methods, including: a high diversity of species to be assessed, the prevalence of multi-species fisheries, and extreme data limitations. Previous stock assessments of US Caribbean resources have attempted to quantify stock status and condition using traditional stock assessment procedures (e.g., yield per recruit (YPR), stock production analyses (ASPIC), catch curve analyses, length frequency examinations). However, nearly all of these evaluations have resulted in an unsatisfactory determination of stock status due to the lack of sufficient data with which to parameterize the models. The Magnuson-Stevens Fishery Conservation and Management Act required the establishment of annual catch limits by 2010 for all stocks, including data-limited stocks. In the absence of sufficient information to conduct traditional stock assessments, catch limits in the U.S. Caribbean have largely been derived from control rules based on recent landings history and are not informed by models accounting for the population dynamics of the species.

In light of these challenges, a previous SEDAR evaluation (SEDAR 46) conducted in 2016 explored the use of a data-limited modeling framework to provide management advice for U.S. Caribbean resources. Due to time constraints, six species were selected for evaluation; this selection was based on a previous data triage exercise conducted by the SEFSC. The stock evaluations conducted for SEDAR 46 explored the use of an analytical process, the Data-Limited Methods Toolkit, which focuses on the development of management advice for data-limited fisheries stocks through the application of data-limited stock assessment models and management procedures. The review phase of this work highlighted a number of important data limitations and assumptions that needed vetting; it was concluded that the approaches could have broad potential use in the Caribbean but that further analysis and more consideration was needed to determine the most ideal applications of the tool to management.

Presently, there are 54 stocks combined in the three U.S. Caribbean Island-based Fishery Management plans, and only four have annual catch limits that are derived from a population dynamics model; the other 50 are still managed using a default average landings control rule. With over a decade of committed effort to produce science-based catch advice in the region, and given the current overall trend of decreasing resource levels for conducting science and assessments, it is clear that the levels of capacity cannot keep up with the demand for more informed management of the full suite of managed species. Particularly under a restricted future budget climate, it is necessary to consider more holistic approaches toward management to address the vast number of species that remain unassessed. Furthermore, the recent thinking and scientific advances in the arena of ecosystem-based fisheries management present new opportunities to apply tools to the U.S. Caribbean that have not previously been considered. The movement toward ecosystem-based fisheries management approaches has also spurred increased consideration of the human dimensions of fisheries management, and offers an opportunity to bridge the gap between scientists and stakeholders in discussions of management frameworks that might be perceived as compatible with the region's goals. The purpose of this SEDAR is to bring this suite of tools, as well as potential tools that have yet to be developed, for consideration in application to U.S. Caribbean fisheries management.

Structure and outcome

This workshop will build on a substantial body of published literature, as well as expertise from the region and from colleagues who have undergone similar challenges (e.g. from the Pacific Islands region). Successful completion of the project objectives will require advance planning and exchange of ideas, well prior to the in-person workshop, to allow for efficient use of the time together. A steering committee has been assembled with members from the SEFSC, SERO, Council and academic partners. In addition to drafting the present document and Terms of Reference for the SEDAR, the steering committee will identify a preliminary suite of analytical approaches and representative case studies to be explored during the workshop. Based on the initial decisions of the steering committee, workshop participants would be identified that would likely include a mix of statisticians and quantitative ecologists, assessment scientists, and social scientists most familiar with the region and its assessment and management challenges.

The outcome of this workshop will be a report documenting recommendations of the workshop participants. This report would include a review of the relevant tools and approaches that were considered, pros and cons of each approach, alignment with present and future expected data availability, and feasibility with respect to legal mandates. It is unlikely that a single best approach will be identified; instead, the report will identify key issues that need to be accounted for along with advantages and disadvantages of the different approaches and feasibility of resolving these issues. Ultimately the results from this workshop will be heavily socialized with the Council with the expectation that the primary identified approaches could be explored for implementation in the region in the near term.

Format and participants

To best accomplish workshop objectives, we propose convening a three-day in-person workshop to review and evaluate potential approaches to addressing defined analytical issues. At least two pre-workshop webinars will likely be required to identify potential analytical approaches and develop a list of analytical tasks and responsible parties; due to the complexities of some of the analyses being evaluated, these webinars will likely need to occur well in advance (3 – 6 months) of the in-person workshop. Following the in-person workshop, at least one post-workshop webinar will be required to tie up any loose ends from the in-person workshop and continue remaining work on the final workshop report. Overall format and content of the webinars and the in-person workshop will be determined by the workshop planning or steering committee, which will consist of a subset of necessary workshop attendees: SEFSC Caribbean Fisheries Branch, SERO Caribbean Branch, SEFSC other representatives, Caribbean territorial representatives, Academic faculty, SSC representatives, and SEDAR coordinators and facilitators.

Table 1. List of relevant literature on data appropriate methods used to inform global fisheries management.

Title	*Citation*	*Approach*
SEDAR Procedural Workshop Caribbean Data Evaluation I	Medley, 2009	Bayesian
Bright spots among the world's coral reefs	Cinner et al., 2016	Bayesian
A composite fishing index to support the monitoring and sustainable management of world fisheries	Ye & Link, 2023	Composite Index
Ecosystem processes are rarely included in tactical fisheries management	Skern-Mauritzen et al., 2016	EAFM
Toward shared understandings of ecosystem-based fisheries management among fishery management councils and stakeholders in the U.S. Mid-Atlantic and New England regions	Biedron & Knuth, 2016	EBFM
Maritime ecosystem-based management in practice: Lessons learned from the application of a generic spatial planning framework in Europe	Buhl-Mortensen et al., 2017	EBFM
Towards ecosystem-based fisheries management in Norway – Practical tools for keeping track of relevant issues and prioritising management efforts	Gullestad et al., 2017	EBFM
Quantifying Patterns of Change in Marine Ecosystem Response to Multiple Pressures	Large et al., 2015	EBFM
Science in support of ecosystem-based management for the US West Coast and beyond	Lester et al., 2010	EBFM
Integrating what? Levels of marine ecosystem-based assessment and management	Link & Browman, 2014	EBFM
Characterizing and comparing marine fisheries ecosystems in the United States: determinants of success in moving toward ecosystem-based fisheries management	Link & Marshak, 2019	EBFM
Ecosystem-Based Fisheries Management for Social–Ecological Systems: Renewing the Focus in the United States with Next Generation Fishery Ecosystem Plans	Marshall et al., 2017	EBFM
Ecosystem-level reference points: Moving toward ecosystem-based fisheries management	Morrison et al., 2024	EBFM
An evaluation of progress in implementing ecosystem-based management of fisheries in 33 countries	Pitcher et al., 2009	EBFM
Evaluating the theoretical and practical linkages between ecosystem-based fisheries management and fisheries co-management	Cucuzza et al., 2020	EBFM and Co-Management

Meeting fisheries, ecosystem function, and biodiversity goals in a human-dominated world	Cinner et al., 2020	Economic Gravity Model
An Integrated Approach Is Needed for Ecosystem Based Fisheries Management: Insights from Ecosystem-Level Management Strategy Evaluation	Fulton et al., 2014	Ecosystem Level MSE, EBFM
Evidence of ecosystem overfishing in U.S. large marine ecosystems	Link, 2021	Ecosystem Overfishing Indicators
Comparative production of fisheries yields and ecosystem overfishing in African Large Marine Ecosystems	Link et al., 2020	Ecosystem Overfishing Indicators
Global ecosystem overfishing: Clear delineation within real limits to production	Link & Watson, 2019	Fogarty, Friedland, and Ryther indices
Gravity of human impacts mediates coral reef conservation gains	Cinner et al., 2018	Gravity Index
Challenges in the assessment and management of small-scale fisheries in Latin America and the Caribbean	Salas et al., 2007	Input control management tools
Evaluating Atlantic bluefin tuna harvest strategies that use conventional genetic tagging data	Carruthers et al., 2023	MSE
Management strategy evaluation of a multi-indicator adaptive framework for data-limited fisheries management	Harford et al., 2016	MSE
Management Strategy Evaluation for Fisheries: Informing the selection of harvest strategies	PEW Charitable Trusts, 2016	MSE
An indicator-based adaptive management framework and its development for data-limited fisheries in Belize	McDonald et al., 2017	Multi-Indicator Framework
Stock Complexes for Fisheries Management in the Gulf of Mexico	Farmer et al., 2016	Multispecies
Multispecies Extensions to a Nonequilibrium Length-Based Mortality Estimator	Huynh et al., 2017	Multispecies
Detecting Multispecific Patterns in the Catch Composition of a Fisheries-Independent Longline Survey	Niella et al., 2017	Multispecies
Discriminating Catch Composition and Fishing Modes in an Artisanal Multispecies Fishery	Purcell et al., 2018	Multispecies
Size structure and gear selectivity of target species in the multispecies multigear fishery of the Kenyan South Coast	Tuda et al., 2016	Multispecies
Sustainable reference points for multispecies coral reef fisheries	Zamborain-Mason et	Multispecies

	al., 2023	
Caribbean Fisheries Data Evaluation SEDAR Procedures Workshop 3	SEDAR, 2009	Multispecies, Multinomial CPUE
A spatial multivariate approach to understand what controls species catch composition in small-scale fisheries	Pennino et al., 2016	Multivariate / SIMPER
Identifying fisheries operations in tropical multispecies fisheries: A comparative analysis of multivariate approaches and neural networks	Quijano et al., 2023	Multivariate Analysis and Neural Networks
Primary production ultimately limits fisheries economic performance	Marshak & Link, 2021	Primary Production
Exploring Tools for Managing Data-Poor Stocks	Caribbean Fishery Management Council and the Fisheries Leadership & Sustainability Forum, 2011	SPR, Density Ratio Control Rule, ERAEF

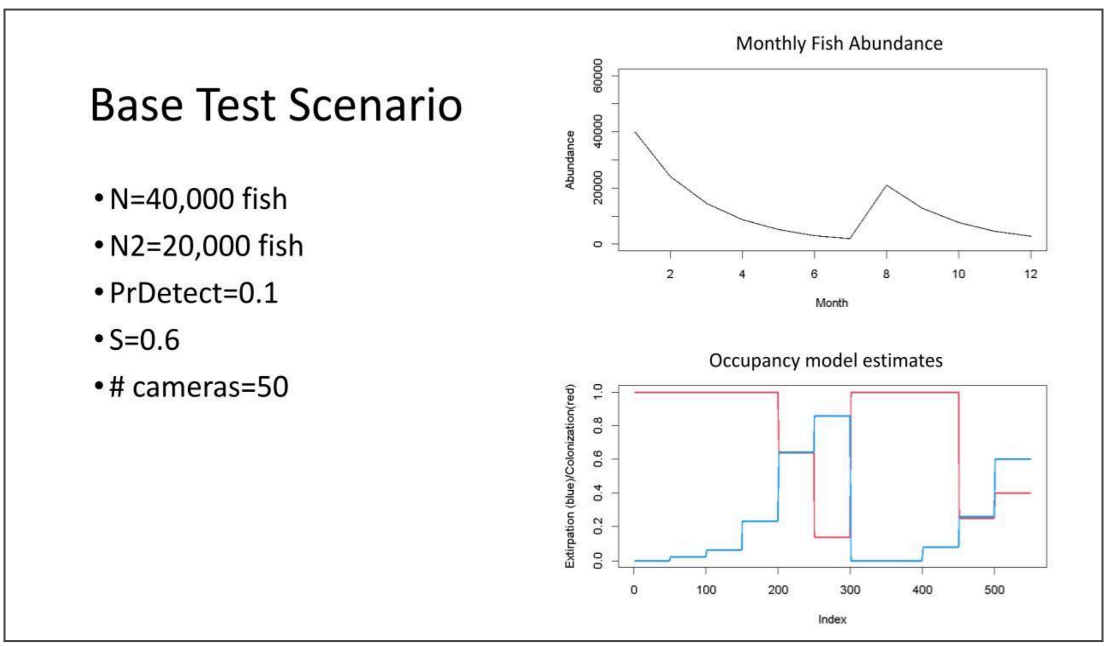

Figure 1. Model estimates for the base scenario.

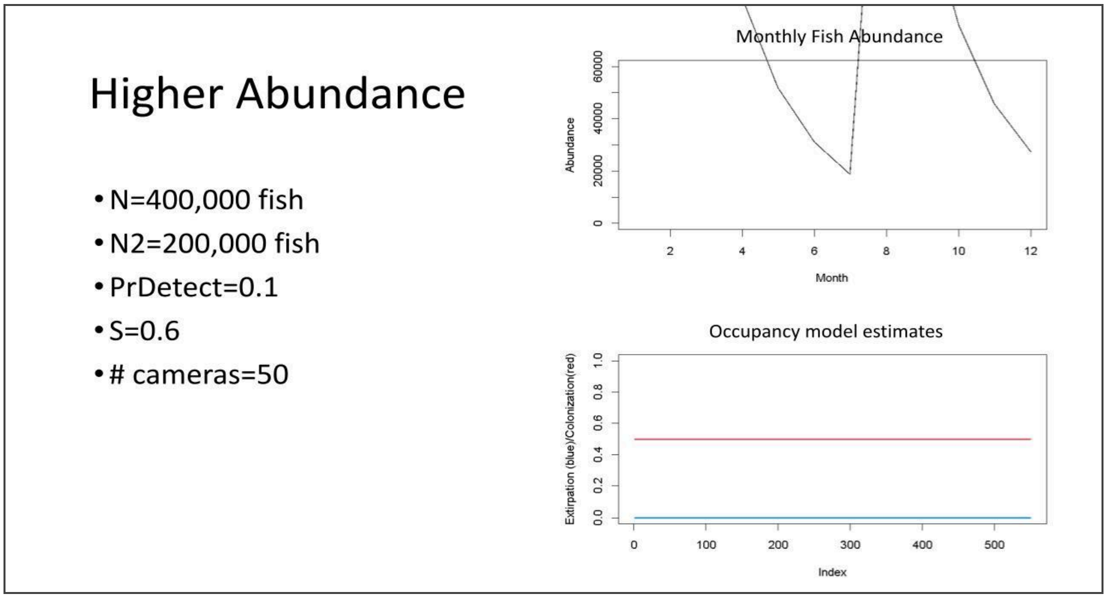

Figure 2. Model estimates for the higher abundance scenario.

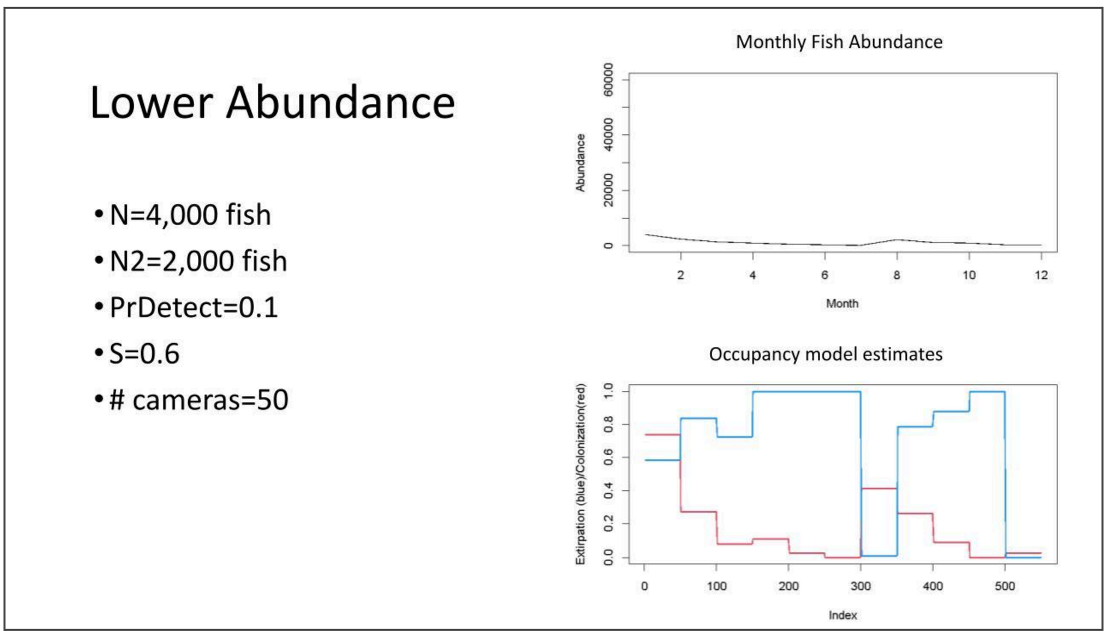

Figure 3. Model estimates for the lower abundance scenario.

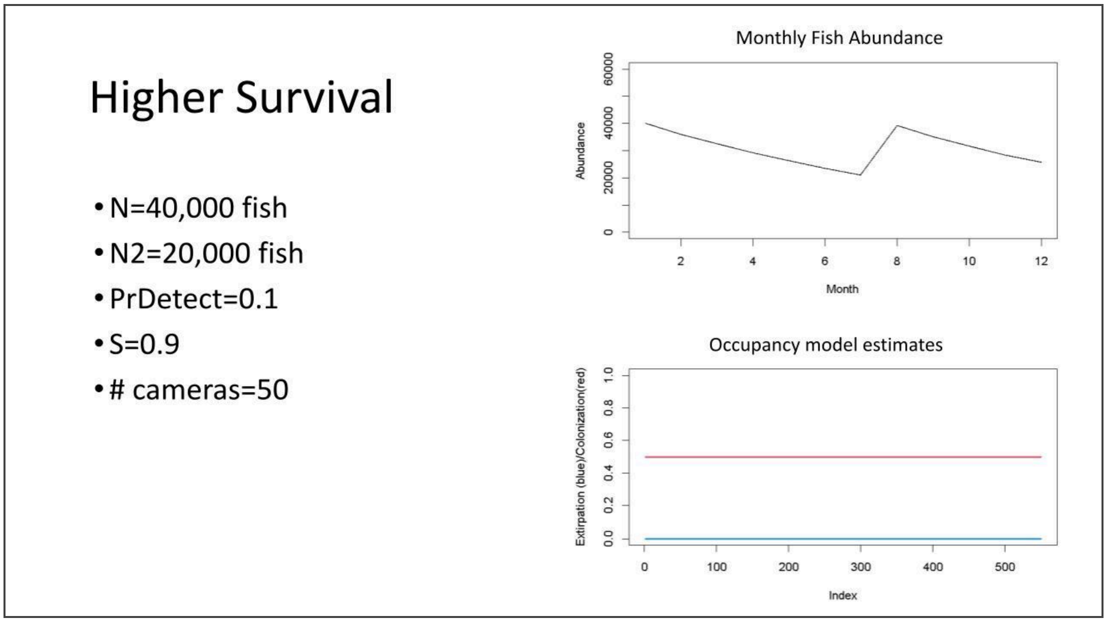

Figure 4. Model estimates for the higher survival scenario.

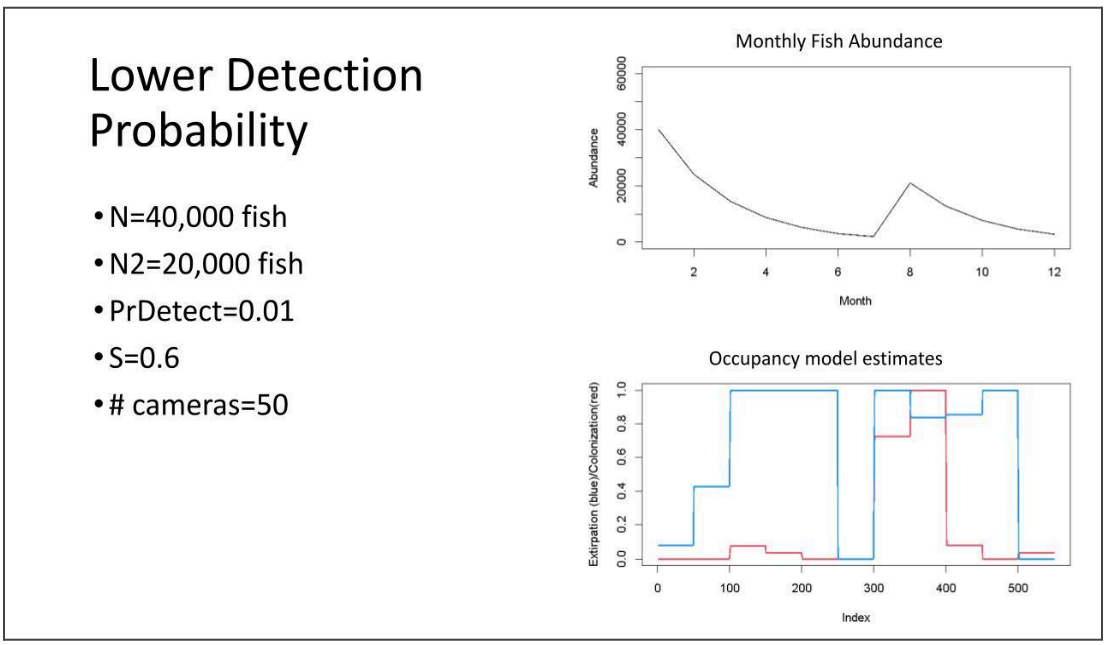

Figure 5. Model estimates for the lower detection probability scenario.

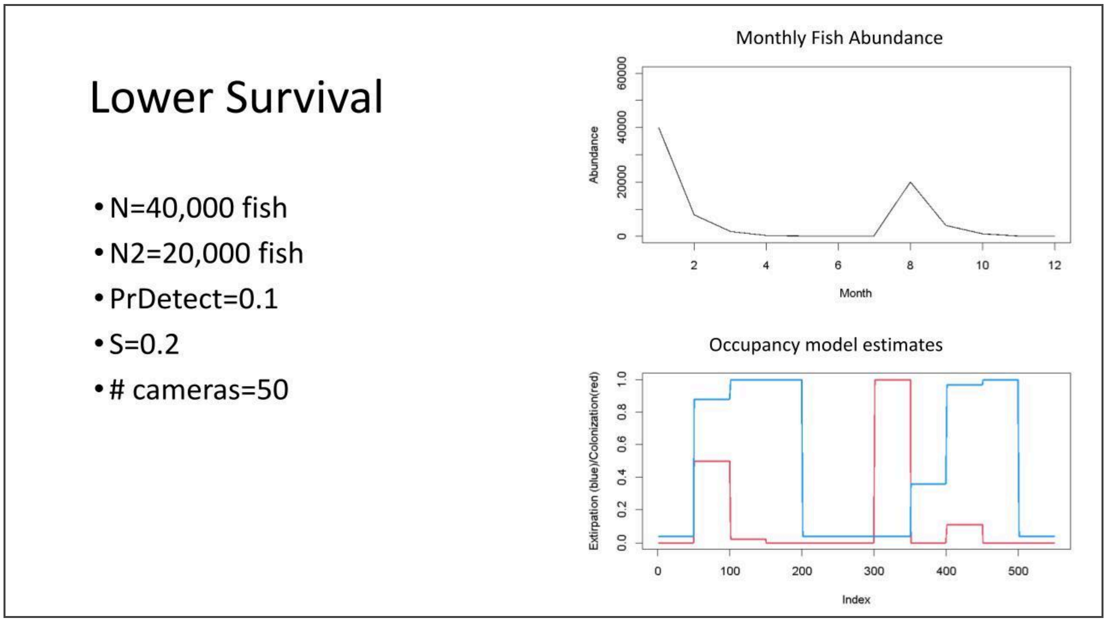

Figure 6. Model estimates for the lower survival scenario.

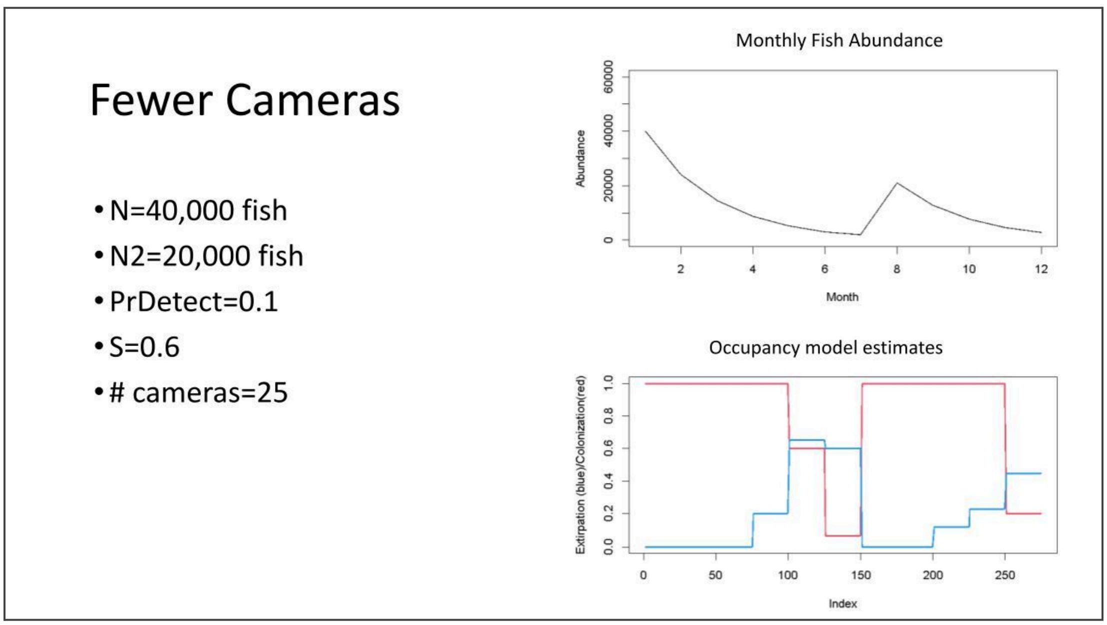

Figure 7. Model estimates for the fewer cameras scenario.

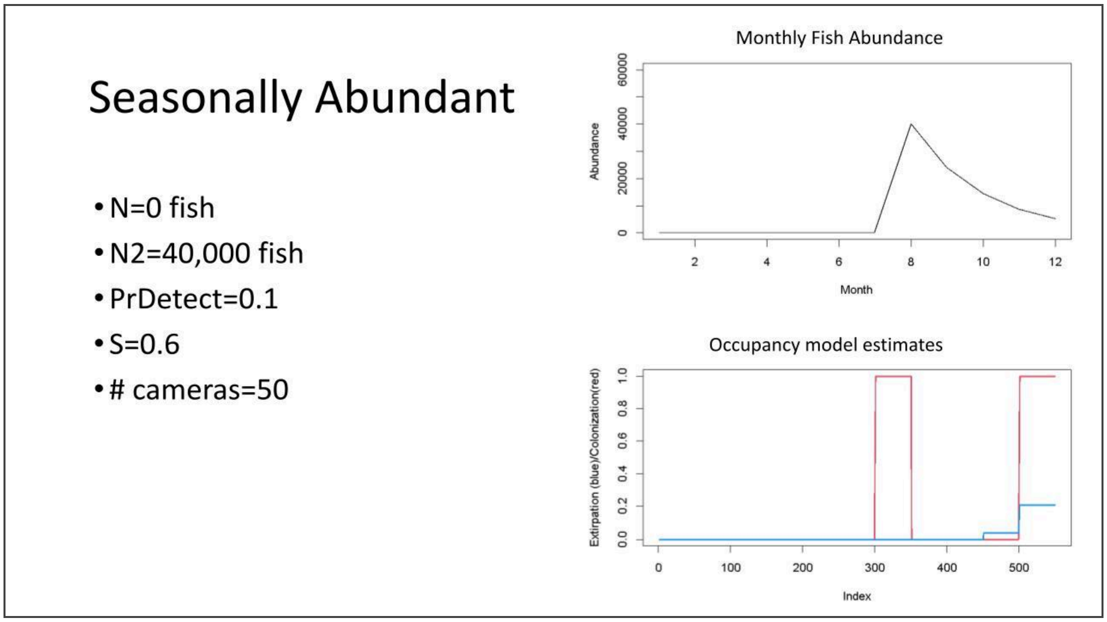

Figure 8. Model estimates for the seasonal abundance scenario.

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